

RESEARCH ARTICLE

Magnetic Resonance in Medicine

Automated generation of cerebral blood flow and arterial transit time maps from multiple delay arterial spin-labeled MRI

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Purpose: Develop and evaluate a deep learning approach to estimate cerebral blood flow (CBF) and arterial transit time (ATT) from multiple post-labeling delay (PLD) ASL MRI.

Methods: ASL MRI were acquired with 6 PLDs on a 1.5T or 3T GE system in adults with and without cognitive impairment ($N = 99$). Voxel-level CBF and ATT maps were quantified by training models with distinct convolutional neural network architectures: (1) convolutional neural network (CNN) and (2) U-Net. Models were trained and compared via 5-fold cross validation. Performance was evaluated using mean absolute error (MAE). Model outputs were trained on and compared against a reference ASL model fitting after data cleaning. Minimally processed ASL data served as another benchmark. Model output uncertainty was estimated using Monte Carlo dropout. The better-performing neural network was subsequently re-trained on inputs with missing PLDs to investigate generalizability to different PLD schedules.

Results: Relative to the CNN, the U-Net yielded lower MAE on training data. On test data, the U-Net MAE was 8.4 ± 1.4 mL/100 g/min for CBF and 0.22 ± 0.09 s for ATT. A significant association was observed between MAE and Monte Carlo dropout-based uncertainty estimates. Neural network performance remained stable despite systematically reducing the number of input images (i.e., up to 3 missing PLD images). Mean processing time was 10.77 s for the U-Net neural network compared to 10 min 41 s for the reference pipeline.

Conclusion: It is feasible to generate CBF and ATT maps from 1.5T and 3T multi-PLD ASL MRI with a fast deep learning image-generation approach.

KEYWORDS

arterial spin labeling, cerebral blood flow, convolutional neural network, deep learning, transit time

1 | INTRODUCTION

Arterial spin-labeled (ASL) MRI is becoming increasingly popular and available for examinations of cerebral blood flow (CBF) in dementia, stroke, and other neurologic and psychiatric diseases.¹ Although most CBF imaging modalities require an injected contrast agent, ASL is non-invasive and, therefore, uniquely suited for follow-up scans and vulnerable patient populations.

The rising availability of ASL in the clinic can be partially attributed to improvements in pulse sequence design of both single post-label delay (PLD) and, more recently, multi-PLD ASL.² Multi-PLD ASL is desirable for clinical application because it enables imaging along different stages of the vascular network, thereby capturing dynamic information about inflowing blood.³ Such information offers the opportunity to quantify the arterial transit time (ATT) and mitigate the transit delay confounds evident in single-PLD ASL, as observed in clinical studies.^{4–6} Multiple PLDs, however, require additional model-based post-processing to synthesize CBF and ATT images—in addition to typical quality controlling measures such as motion correction. The prevailing model-fitting approaches are computationally cumbersome and time consuming, thereby limiting the clinical adoption of multi-PLD ASL.⁷

Deep learning offers advantages for many stages of conventional image processing pipelines,⁸ as is the case for de-noising and artifact removal in single-PLD ASL processing.^{9–12} Deep neural networks have also been advantageous in de-noising Hadamard-encoded ASL images and as parameter estimation models in ASL MR fingerprinting applications.^{13–16} These developments suggest the feasibility of deep neural networks for de-noising and hemodynamic parameter quantification of sequential multi-PLD ASL. Some ongoing challenges of deep learning for ASL processing include 3 dimensional (3D) architectures, uncertainty in network outputs, generalizability to acquisition sequence, and diverse clinical populations.

In this study, we aimed to demonstrate that a 3D deep learning-based processing pipeline is feasible for fast and accurate estimation of entire CBF and ATT image volumes from multi-PLD ASL acquired on 2 scanners. We performed 3 experiments using scans of older participants with and without cognitive impairment, acquired at either 1.5T or 3T. First, we trained 2 distinct 3D convolutional neural networks using each PLD as a separate input channel and performed cross-validation to select the better-performing architecture. Next, we examined the test-set performance of the better-performing network and estimated the association between a network uncertainty heuristic and error. Finally, we investigated whether an

alternative training scheme could improve model performance when tested on inputs with fewer PLD images, emulating alternative PLD schedules.

2 | METHODS

2.1 | Participants

This is a retrospective study of 99 participants scanned at Sunnybrook Research Institute in Toronto, Ontario (ON). Research ethics approval was granted by the ethics board at Sunnybrook Research Board. Participants were recruited through a neurology clinic at Sunnybrook Health Sciences Centre (detailed inclusion/exclusion criteria can be found at <https://clinicaltrials.gov/ct2/show/NCT01800214>). Participants were either cognitively unimpaired, had a mild cognitive impairment diagnosis, or had a dementia diagnosis.

2.2 | Magnetic resonance imaging

Imaging was performed on a 1.5T GE Signa scanner (General Electric Medical Systems, Milwaukee, WI) for 50 participants and on a 3T GE Discovery 750 scanner for 49 participants. Both scanners were equipped with an 8-channel head array coil for signal reception. As detailed in the main text, the MRI protocol consisted of ASL and anatomic imaging, namely T₁-weighted images. ASL MRI was nearly identical on the 2 scanners and included a 2D echo planar imaging acquisition, 64 × 64 × 17 matrix, 3.5 × 3.5 × 5.6 mm³ voxel resolution, TR/TE = 3500–5000/17 ms, pseudo-continuous labeling (1500 ms label duration, 6 PLDs of 100, 500, 900, 1300, 1700, and 2100 ms collected in a sequential, non-interleaved fashion), and 12 control-label image pairs for each PLD. Labeling was performed using 1000 504 μs Hanning-shaped 35° pulses. The labeling plane was 5-mm thick and was positioned immediately superior to the carotid bifurcation and perpendicular to the internal carotid arteries. Total scan time per PLD was 24 × TR = 84–120 s. Additional ASL sequence details have been previously reported.¹⁷ An M₀ image for CBF quantification was also acquired at the beginning of each different PLD. Anatomic T₁-weighted imaging was performed for registration and tissue segmentation. The anatomic 1.5T T₁-weighted acquisition was an axial 3D SPGR with TR/TE = 35/5 ms, flip angle 35°, 256 × 256 matrix, 0.9 × 0.9 mm² resolution, and 1.2–1.4 mm slice thickness, depending on participant head size. The 3T T₁-weighted acquisition was an axial 3D FSPGR with

TR/TE/TI: 8.1/3.2/650 ms, flip angle 8°, 256 × 256 matrix, 0.86 × 0.86 mm² resolution, and 1.0 mm slice thickness.

2.3 | Image processing

2.3.1 | Reference pipeline

Reference parameter maps used as ground truth for model training were estimated using a standard ASL MRI processing pipeline built with tools from the FMRIB Software Library (FSL). Motion correction was applied by registering the time series to the middle volume.¹⁸ CBF-weighted difference images were calculated via sinc subtraction of control and label volumes.¹⁹ Smoothing was applied with a Gaussian kernel with 5 mm full-width at half maximum (the same smoothing was applied to the long-TR reference image). Motion-corrupted difference images were discarded using the ENABLE quality control algorithm.^{20,21} Images were skull-stripped by applying a brain mask estimated from segmentation of the T₁-weighted image. CBF and ATT were estimated using the BASIL software included with FSL, which incorporates ASL data from all acquired PLDs into an inference scheme based on the single-compartment kinetic model established by Buxton et al.^{22,23} Model parameters were chosen following the community consensus paper by Alsop et al.⁷ Resultant ATT images were in units of seconds and CBF images estimated by BASIL were converted to units of mL/100 g/min using a standard calibration equation.⁷

2.3.2 | Proposed pipeline

Figure 1A shows the proposed pipeline schematic. First, difference images were calculated with sinc subtraction of raw control-label ASL volumes. The resultant images were averaged at each PLD, resulting in a 6-channel input to the neural network. Input and reference output image intensities were standardized by dividing each voxel by the 95th percentile of intensities in the entire training sample. CBF and ATT images were standardized separately. The network outputs were designed to be standardized CBF and ATT images that were subsequently re-scaled using the inverse of the standardization transform (i.e., multiplied by the 95th percentile of voxel intensities).

2.4 | Network implementations

We implemented 2 distinct convolutional neural network architectures in Python 3.5 using the Keras API of the TensorFlow 2.2 library. As shown in Figure 1B, the 2

architectures were: (1) a 3D convolutional neural network (CNN) and (2) a U-Net.²⁴ In both cases, trainable parameters were initialized with normally distributed random values.

For comparison of the 2 network architectures, the same 6-image inputs were used (i.e., one 3D image per PLD), along with 2 output images (i.e., one 3D map for each of CBF and ATT). By design, each model varied in the number of trainable parameters.

The CNN consisted of five 3D convolutional layers with a 3 × 3 × 3 kernel followed by a ReLU activation function. The U-Net included 7 convolutional layers with a 3 × 3 × 3 kernel, 3 of which downsampled the image resolution (i.e., U-Net depth of 3) and 3 upsampled the images back to the original resolution. Down and upsampling was performed by a factor of 2 in the x-y plane using convolutions with stride 2.

2.5 | Training details

Both the CNN model and the U-Net model were trained on the same data, that is, a mix of 1.5T and 3T images. We augmented each image (all input and output images) in the training set by applying 3 separate transformations: (1) mirroring in the x-y plane, (2) rotating by +20° in the x-y plane, and (3) rotating by −20° in the x-y plane. Therefore, the total dataset used for cross-validation was 316 image sets (comprising the original 79 times 3 augmentations for each; 40 images were acquired at 1.5T and 39 at 3T).

Training of both networks was performed on a Linux system with an NVIDIA P100 Pascal GPU (NVIDIA, Santa Clara, CA). For the CNN and U-Net, the loss function was the mean absolute error (MAE) adjusted to assign equal weighting to brain and background voxels, defined using the average of all images in the training set. Explicitly, the loss function was,

$$\mathcal{L}(y, C(x)) = \frac{\sum_{i=1}^n |y_i - C(x)_i|}{2n} + \frac{\sum_{j=1}^m |y_j - C(x)_j|}{2m}, \quad (1)$$

where y is the ground truth reference image, C is the neural network (CNN or U-Net), x is the input image, n is the number of brain voxels, and m is the number of background voxels. We used the Adam optimizer with an initial learning rate of 0.0005, which was reduced to 0.0001 after 100 epochs. Visual inspection of the loss curves was used to ensure plateau of the training and validation losses and to avoid overfitting. Each model was trained for 200 epochs during cross validation using a batch size of 5 image sets. The best performing model was trained for 600 epochs on the entire training set before testing, also using a batch size

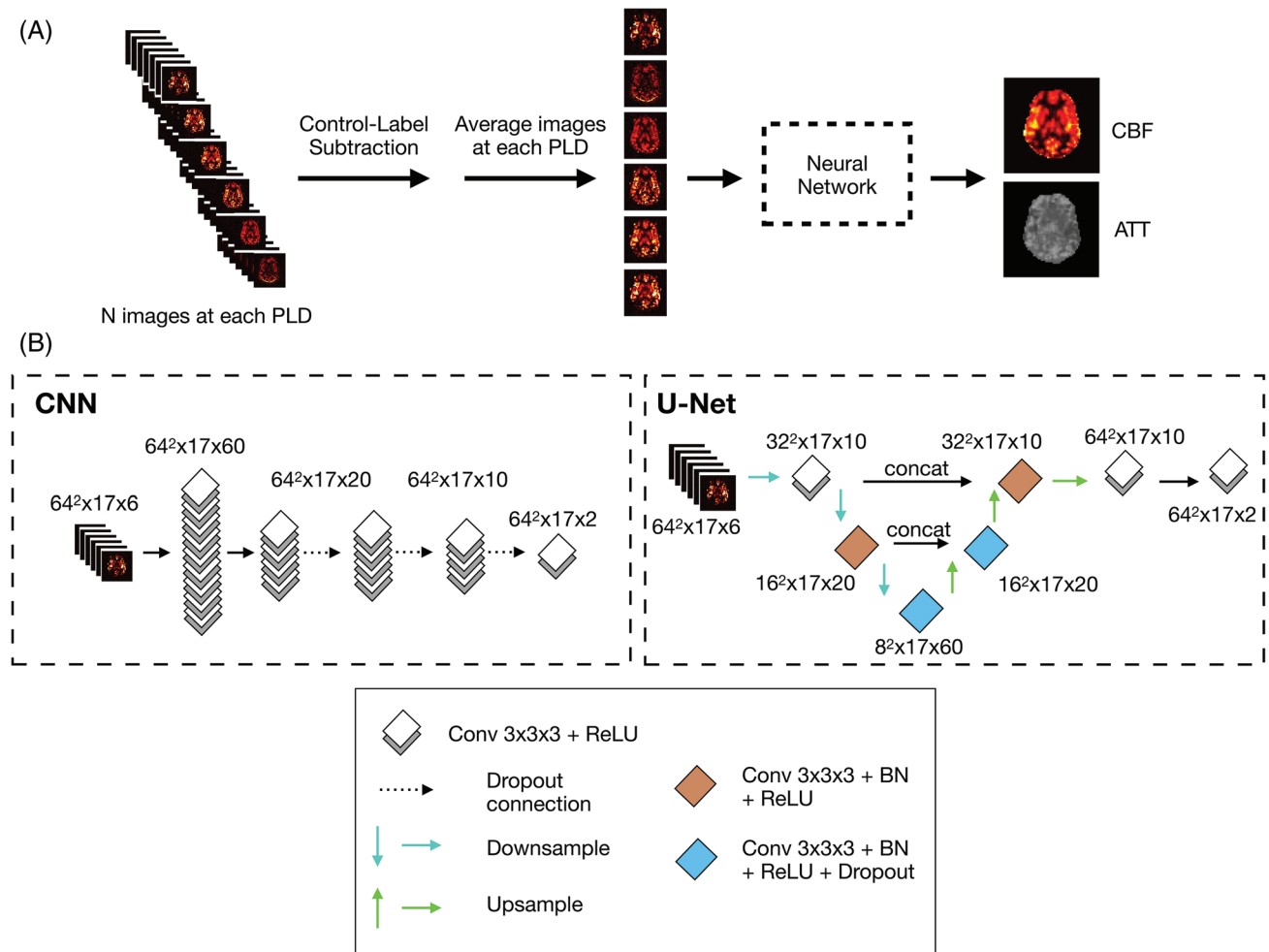


FIGURE 1 (A) Image processing pipeline. (B) Network architectures. Architectures are shown in dashed-line boxes, whereas the legend is shown in the solid-line box. BN, batch normalization; PLD, post-label delay; ReLU, rectified linear unit

of 5. The number of trainable parameters in each of the networks was: CNN = 58 592; U-Net = 88 792.

The best performing model architecture was selected based on the lowest aggregate (normalized CBF and ATT) MAE of gray matter voxels across all validation data in the cross validation.

2.6 | Experiments

Before cross-validation, a random 80%/20% train/test split was used. Splits were stratified by scanner magnetic field strength such that there was the same proportion of 3T and 1.5T scans in both the training and testing splits.

2.6.1 | Cross-validation for architecture selection

A 5-fold cross-validation of the training set, stratified by scanner magnetic field strength, was used to select the

best-performing network architecture. Performance was evaluated based on aggregate (CBF and ATT) MAE of gray matter voxels.

2.6.2 | Evaluation of model performance and model uncertainty estimation

We re-trained the architecture corresponding to the best-performing model on the entire training set. Performance was evaluated on the test set using MAE and the structural similarity (SSIM) index for both CBF and ATT. MAE was evaluated across all gray matter, all white matter, and 5 gray matter regions: frontal lobe, parietal lobe, occipital lobe, temporal lobe, and subcortical gray matter. Regions were identified using the MNI152 atlas and registered to each individual's ASL image using FSL FMRIB's Linear Image Registration Tool (FLIRT), and using registration to the individual's T_1 -weighted image as an intermediate step.²⁵

We used dropout (random deletion of network connections) during testing to produce non-deterministic network estimates.²⁶ The dropout rate was set to 0.30. Two uncertainty heuristics were subsequently considered: standard deviation and coefficient of variation across 500 samples.

2.6.3 | Generalizing to cases with missing input images via curriculum training

We re-trained the model using a learning scheme, in which 1 or more of the 6 input images were deliberately missing (i.e., image intensities set to 0). This strategy emulates changing the PLD schedule and reducing the scan duration. Examples with missing data were introduced during training in order of increasing difficulty, meaning more inputs missing and, therefore, less information provided (a strategy known as curriculum training).²⁷ We removed inputs in a systematic outside-in fashion: first the 100 ms PLD image was removed, then 2100 ms, then 500 ms.

Curriculum training was performed for 600 total epochs wherein the inputs with missing PLDs were introduced starting at epoch 200, after which the difficulty of training examples was increased (i.e., another PLD was removed) at epochs 300 and 400.

2.7 | Processing time determination

Total processing time of both the proposed pipeline and the reference pipeline were estimated by timing the processing of a single dataset 10 times on the same computer.

2.8 | Performance of BASIL model fitting without quality controlling preprocessing steps

To compare our deep learning approach to a comparably fast method that uses standard model-fitting, we calculated CBF and ATT using the BASIL model fitting software on the same (minimally preprocessed) images used as input for the deep neural networks. We refer to this pipeline as the minimal pipeline because motion correction and the ENABLE algorithm are not applied. As previously described, these images underwent only 2 preprocessing steps: control-label subtraction and averaging across repeated acquisitions for each PLD. We, then, computed the MAE and SSIM metrics between this processing pipeline and the full reference pipeline.

2.9 | Statistical analyses

All data were tested for normality with the Shapiro–Wilk test. In the case of a significant result (a priori significance threshold = 0.05), non-parametric statistical tests were used.

We performed a 2-tailed Wilcoxon signed-rank test to assess differences between the cross-validation performance of the CNN and U-Net. On the test set, we performed a 2-tailed *t* test (a priori significance threshold = 0.05) to test for differences between field strengths in gray matter MAE of CBF and ATT (Wilcoxon signed-rank test was used for ATT because of evidence of non-normality).

The performance (MAE and SSIM) of the U-Net and the minimal pipelines were compared with the Wilcoxon signed-rank test for both CBF and ATT.

The relationship between regional MAE in gray matter and the standard deviation uncertainty heuristic was investigated with a linear mixed-effects model with gray matter region as a random intercept term.

To evaluate whether curriculum training improves performance when 3 PLDs were removed from the input, a 2-tailed Wilcoxon signed-rank test compared MAE of the curriculum-trained model with the model trained on only 6-input images. Using a repeated-measures ANOVA, we tested for differences in CBF MAE of the curriculum-trained model between examples with 0, 1, 2, and 3 input images removed; in the case of ATT, we used the non-parametric equivalent of repeated-measures ANOVA, the Friedman test, because there was evidence for non-normality of ATT MAE.

3 | RESULTS

3.1 | Participant demographics

Demographics and basic clinical information of the 99 participants are shown in Table 1. Scanning was performed at 1.5T for 50 participants and at 3T for 49 participants. Twenty-three participants were cognitively unimpaired, 17 had a mild cognitive impairment diagnosis, and 59 had a dementia diagnosis.

3.2 | Selection of network architecture: cross-validation on training data

The MAE of CBF estimates across 5-fold cross-validation were significantly different between the U-Net and CNN: CNN = 10.3 ± 5.0 mL/100 g/min and U-Net = 9.2 ± 1.7 mL/100 g/min (Wilcoxon signed-rank

TABLE 1 Participant demographics. Additional detailed diagnostic information is provided for the group with a dementia diagnosis

	1.5T images (<i>N</i> = 50)	3T images (<i>N</i> = 49)
Demographics		
Age, mean (SD)	73.1 (8.8)	73.6 (8.1)
Females, <i>n</i> (%)	27 (54.0)	30 (61.2)
Dementia diagnosis		
Alzheimer's disease	15	23
Cortical basal syndrome	3	0
Dementia with Lewy bodies	3	0
Frontotemporal dementia	7	4
Vascular dementia	1	0
Cortical basal syndrome/ progressive supranuclear palsy	1	0
Dementia with Lewy bodies/ Parkinson's disease dementia	1	0
Alzheimer's disease/ frontotemporal dementia	1	0
Mild cognitive impairment	16	1
Cognitively unimpaired	2	21

Abbreviation: SD, standard deviation.

test for difference: $P = 0.010$). The MAE of ATT estimates were: CNN = 0.30 ± 0.18 s and U-Net = 0.27 ± 0.18 s. Because the U-Net produced estimates with the lowest MAE, we chose the U-Net as the best-performing architecture to consider for the next experiments.

3.3 | Test set performance of the U-Net

The performance of the U-Net on the test set is shown in Figure 2 and Table 2. Average (mean $\pm$ standard deviation) gray matter MAE for CBF and ATT estimates was 8.4 ± 1.4 mL/100 g/min and 0.22 ± 0.09 s. Average gray matter MAE was lower for CBF (2-tailed t test: $t = -3.04$, $P = 0.014$), but not ATT (2-tailed Wilcoxon signed-rank test: $W = 25.0$, $P = 0.799$), in the scans acquired at 1.5T compared to 3T. Representative high and low MAE network estimates and corresponding reference images are shown in Figure 3.

Relative to the full reference pipeline, the MAE of the minimal pipeline (i.e., BASIL model fitting on minimally preprocessed images) was 9.88 ± 2.36 mL/100 g/min for gray matter CBF and 0.47 ± 0.17 s for gray matter ATT; in both cases the MAE of the U-Net outputs was lower

(CBF: $P = 0.028$; ATT: $P < 1.0 \times 10^{-4}$). The whole-image SSIM index of the minimal pipeline was 0.69 ± 0.10 for CBF and 0.75 ± 0.03 for ATT; in both cases the average SSIM of the U-Net outputs was higher (CBF: $P < 1.0 \times 10^{-4}$; ATT: $P = 0.017$) (see Table 2 for quantities).

We also tested the trained U-Net on inputs processed using motion correction and the ENABLE outlier-removal algorithm. In comparison to the original experiment that used inputs without these preprocessing steps (Table 2), the preprocessed inputs resulted in better U-Net performance for both gray matter CBF (MAE = 7.66 ± 1.06 mL/100 g/min; SSIM = 0.81 ± 0.04) and ATT (MAE = 0.20 ± 0.04 s; SSIM = 0.79 ± 0.03).

A proxy for input image quality, the mean number of image volumes removed by the ENABLE algorithm was 8.5 volumes (median = 9.5) for the 1.5T test set images and 14.7 volumes (median = 11.5) for the 3T images.

3.4 | Network uncertainty estimation

Figure 4A,B show the network uncertainty heuristics and MAE for CBF and ATT. Across individuals, a linear mixed effects model revealed an association between uncertainty and MAE of both CBF ($\beta = 0.16$, $t = 2.64$, $P = 0.0097$) and ATT ($\beta = 0.048$, $t = 3.43$, $P = 0.0009$). Representative uncertainty maps for high and low-quality CBF estimates are shown in Figure 4C,D, respectively.

3.5 | Curriculum training for parameter estimation with <6 input images

MAE as a function of missing ASL data is shown in Figure 5. Two models are depicted: 1 trained on examples with missing input data (i.e., curriculum-trained) and 1 trained with only the full input data. The curriculum-trained model had significantly lower CBF MAE relative to the model trained on full inputs when considering inputs with 3 images removed (2-tailed Wilcoxon signed-rank test: $W < 0.1$, $P = 8.86 \times 10^{-5}$). Repeated-measures ANOVA revealed no differences in CBF MAE of the curriculum-trained model across the schedules with 0, 1, 2, and 3 input images removed ($F = 1.70$, $P = 0.177$).

For ATT estimates, the curriculum-trained model similarly had significantly lower MAE relative to the model trained using full inputs when considering inputs with 3 images removed (2-tailed t test $t = -5.67$, $P = 1.8 \times 10^{-5}$). There were significant differences in ATT MAE of the curriculum-trained model across the schedules with 0, 1, 2, and 3 input images removed (Friedman test: $F = 19.92$, $P = 0.0002$).

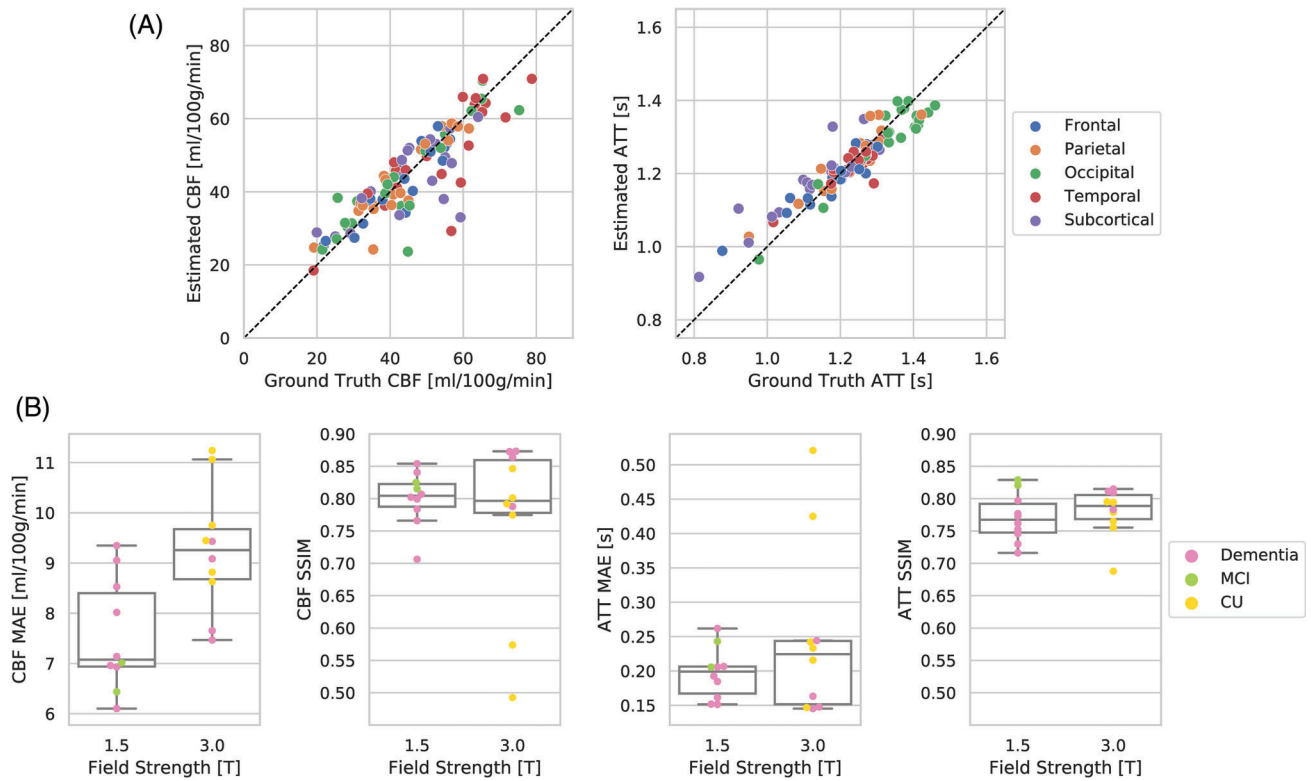


FIGURE 2 Test set performance in gray matter. (A) Association between network estimate and reference images averaged across 5 gray matter regions (resulting in 5 data points per participant). Colors denote gray matter region. (B) Mean absolute error (MAE) and structural similarity (SSIM) index between network estimate and reference images. Samples are stratified by scanner field strength. Colors denote clinical status. CU, cognitively unimpaired; MCI, mild cognitive impairment

TABLE 2 Test set ($n = 20$) performance characteristics. For reference, average reference gray matter CBF and ATT across the test set was 51.1 ± 12.1 mL/100 g/min and 1.21 ± 0.08 s

	CBF (mean $\pm$ SD) (mL/100 g/min)	ATT (mean $\pm$ SD) (s)
Mean absolute error		
Gray matter	8.4 ± 1.4	0.22 ± 0.09
Frontal lobe	8.3 ± 2.1	0.18 ± 0.07
Occipital lobe	8.2 ± 1.5	0.25 ± 0.11
Parietal lobe	7.6 ± 1.4	0.16 ± 0.07
Temporal lobe	9.1 ± 1.8	0.30 ± 0.15
Subcortical	7.5 ± 3.0	0.16 ± 0.13
White matter	7.5 ± 1.9	0.16 ± 0.07
Structural similarity (entire image; unitless)	0.78 ± 0.10	0.77 ± 0.04

Abbreviation: SD, standard deviation.

3.6 | Processing times for the U-Net, reference, and minimal pipelines

The total processing time of the U-Net approach was 10.77 ± 0.19 s (which includes loading and saving images; computing control-label difference images;

and averaging repeat volumes). The forward pass of the U-Net model alone took 2.44 ± 0.03 s. In comparison, the total processing time of the reference calculation was $10 \text{ min } 41 \text{ s} \pm 40.8 \text{ s}$. The total processing time of the minimal pipeline was 14.81 ± 0.37 s.

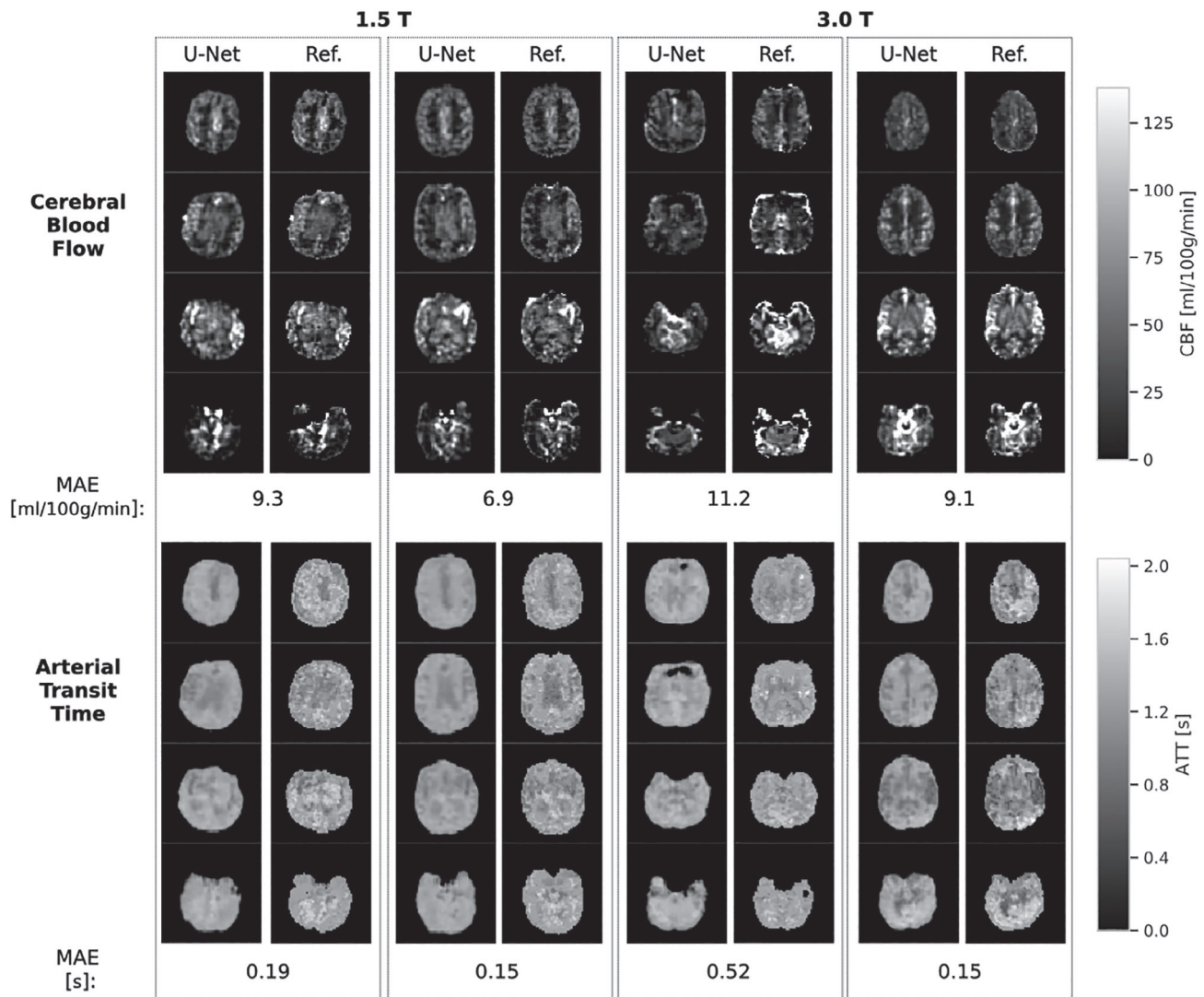


FIGURE 3 Higher-quality and lower-quality examples of CBF and ATT estimates for both 1.5T and 3T scans. Each panel displays 4 axial slices from the same image (top: most superior, bottom: most inferior). Each pair of panels displays the network estimate and the reference image on the left and right, respectively. Each column displays images from the same individual. The lower-quality 3T example shows the lowest performance in the test set and is an outlier based on mean absolute error (N.B. this participant appeared to have the largest amount of head motion; specifically, the ENABLE outlier removal algorithm removed 30 of 72 total volumes). MAE, mean absolute error; Ref., reference image

4 | DISCUSSION AND CONCLUSIONS

In this study, we trained deep CNN to estimate cerebral blood flow and arterial transit time maps from multi-PLD ASL MRI. The U-Net modestly outperformed the standard CNN with average CBF estimation error of 8.4 ± 1.4 mL/100 g/min on the test set. U-Net uncertainty estimates were associated with MAE of the network-generated CBF and ATT maps. Last, through a curriculum learning scheme, we trained the U-Net to improve its performance when fewer image inputs were provided. The U-Net-based pipeline was substantially

faster than the reference pipeline, a crucial consideration in time-constrained clinical settings.

Overall, the average test set CBF error of the U-Net model was comparable to the variability that tends to occur because of different processing choices in a standard ASL pipeline, reported to be between 5–10 mL/100 g/min.²⁸ This finding suggests our proposed pipeline is feasible for CBF estimation and there is a modest data quality penalty despite substantive savings in processing time. As expected, when testing the U-Net with motion-corrected and ENABLE-processed input images, we observed better performance compared to using inputs without these preprocessing steps. Notably, between-subject variability

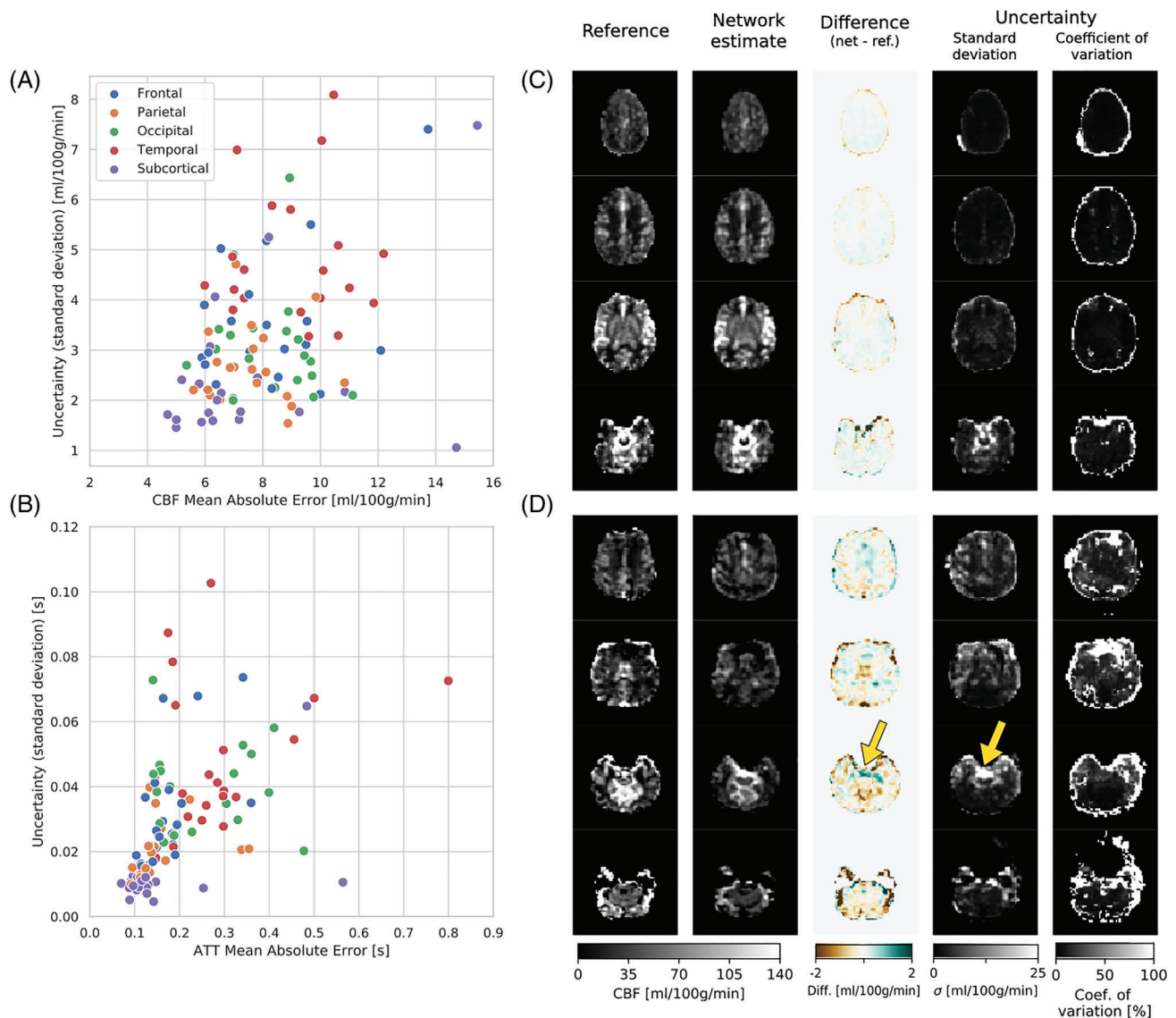


FIGURE 4 Network uncertainty heuristics were estimated through Monte Carlo dropout. Scatterplots in (A) and (B) show the association between network uncertainty and mean absolute error. A shows regional gray matter CBF whereas B shows regional ATT, and colors denote gray matter region. Images shown in (C) and (D) help to visualize network-estimated CBF uncertainty maps for 2 cases acquired at 3T. A higher-quality example is shown in C and a lower-quality example is shown in D. Yellow arrows highlight an example of an area where high uncertainty overlaps with high error. (See Supporting Information Figure S1 for network-estimated uncertainty from cases acquired at 1.5T).

in network performance also dropped, which may suggest reducing outlier features in the inputs (e.g., motion) helps make more consistent network estimates. Future applications may explore the tradeoff between computing time and the addition of quality-controlling preprocessing steps before neural network processing. Indeed, by removing the quality-controlling steps from the reference pipeline and calculating CBF/ATT with the BASIL model fitting, we observed a performance penalty alongside a computational speedup. Despite this speedup, the U-Net approach was faster and demonstrated better performance.

The U-Net outperformance of the CNN during cross-validation was significant, but modest. Although this may be a consequence of the U-Net's more sophisticated architecture design (with encoder and decoder network arms) or increased depth, it may also be sample-specific or related to the number of trainable parameters. More work on this comparison is warranted.

Both the U-Net and the CNN were trained on mixed data from both 1.5T and 3T scanners because of the need for a model that is agnostic to field strength. With larger datasets, it would be interesting to train neural networks for 1.5T and 3T separately, to compare with

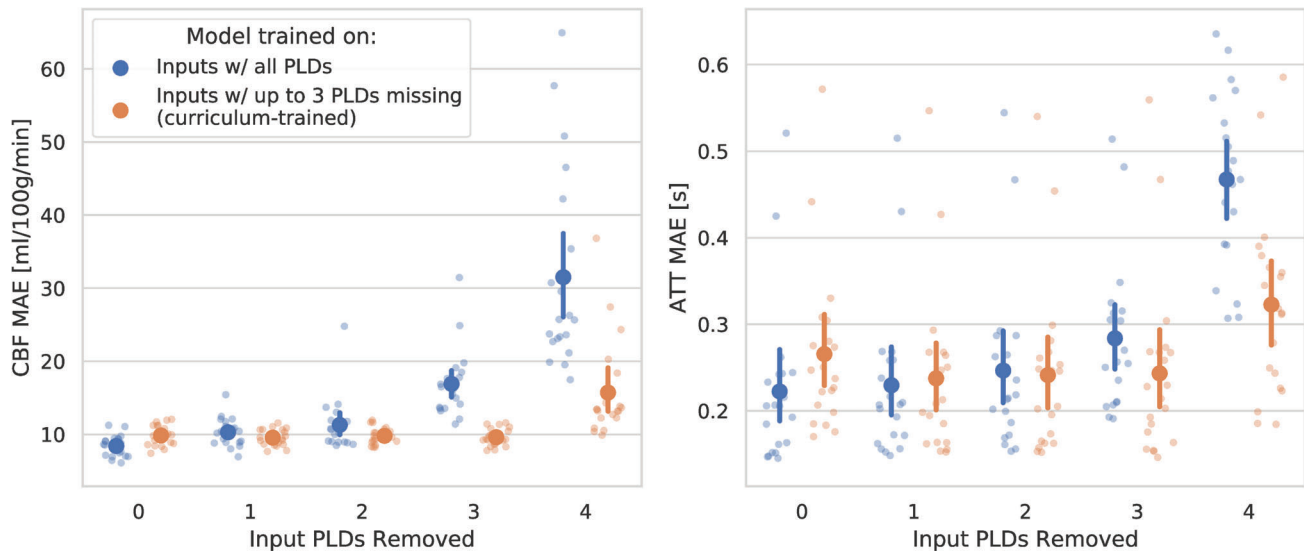


FIGURE 5 Model performance when PLDs are removed from the input, emulating a different PLD schedule. Performance of 2 models is shown: (1) the model trained on examples with all 6 images as input; (2) the curriculum-trained model, trained on examples with up to 3 input images missing. Testing on examples with up to 4 input images missing is visualized for context. Translucent points show individuals' data, large solid circles show mean, and bars show 95% confidence intervals. MAE, mean absolute error; PLD, post-label delay

the U-Net of the current study, and to test whether ASL field-strength-dependent information can be predicted (e.g., T_1 of arterial blood). In the present study, we investigated performance differences of the U-Net between estimates from 1.5T and 3T data. In the test set, we observed a significant effect of scanner field strength on network CBF MAE; somewhat unexpectedly, 1.5T data had lower MAE than 3T data, which may reflect 2 considerations. First, the participant demographics were different for the 2 scanners, with more cognitively unimpaired individuals scanned at 3T. Second, the 1.5T test set images may have been less impacted by motion than the 3T images, as evidenced by the higher number of 3T image volumes removed by the ENABLE algorithm. We observed 2 outliers in this 3T cohort, based on low SSIM index. These outliers were less prominent based on the MAE metric, which may reflect both the choice of MAE as the training loss function and the fundamental differences between MAE, which assesses differences between each voxel, and SSIM, which is a whole-image summary metric.

Estimating model uncertainty provided insight on regional model performance without the need for reference comparison. Access to uncertainty information is pertinent in clinical deployment of deep learning methods: uncertainty can offer quantification of image quality and is a potential avenue for automated quality control. Indeed, the standard deviation-based uncertainty heuristic was associated with both CBF and ATT errors in gray matter. As a secondary metric, we also evaluated uncertainty based on the coefficient of variation,

which displayed qualitatively distinct contrast relative to standard deviation. This was likely because of association between standard deviation and CBF; therefore, the coefficient of variation, which is the standard deviation relative to CBF, may be differently sensitive to errors.

By training the network on inputs with missing PLDs from the 6-PLD acquisition, we demonstrated preliminary evidence that the network may generalize to accommodate different acquisitions, which is desirable to allow patient-specific PLD schedules and to reduce scan duration. As expected, the model performed better on 3-PLD inputs after it trained on examples with a 3-PLD input. Notably, this curriculum-trained model exhibited no performance decrements in CBF estimates across inputs with up to 3 missing PLDs. Therefore, without the need for re-training, this model may be used to process images acquired with different PLD schedules; in the current implementation, the different schedules must be a subset of the schedule on which the model was trained, as would be the case in time-constrained or motion-corrupted imaging scenarios. Conversely, considering ATT estimates, the curriculum-trained model performed better using inputs with 3 PLDs removed compared to inputs with fewer PLDs removed. For both CBF and ATT, curriculum training protected against an MAE increase when PLDs were systematically removed during testing. Further research may be warranted to advance network performance and robustness to missing data, which could include refinements to the learning scheme and

modifying the parameter estimation method used for reference, particularly ATT (e.g., the method developed by Dai et al.²⁹).

Study limitations are discussed herein. First, the models do not explicitly incorporate tissue equilibrium magnetization, which is vital for absolute quantification of CBF. In the current study, the model may learn to scale appropriately because there are only 2 scanners; generalizability would require additional investigation of a broader set of pulse sequences and scanners. Second, although we implemented 2 network architectures, we did not perform exhaustive hyperparameter tuning. Therefore, the results of the current study provide a benchmark for future studies, in which hyperparameters such as network architecture and alternative loss functions (including the relative brain and background weighting) can be tested. Third, the cognitive neurology sample was relatively small, which limits the investigation of sources of performance variability and sensitivity to diagnosis. Fourth, we use a single-compartment ASL kinetic model for generation of the ground truth reference images. Additional studies may focus on, for example, network estimation of arterial blood volume using a 2-compartment kinetic model. Finally, the imaging sequence studied here is 1 implementation of multi-PLD ASL and is longer than would be ideal for clinical implementation; in the absence of a standardized protocol, future work is needed to expand on pulse sequence parameters.

In conclusion, we demonstrated the feasibility of using a deep neural network to estimate CBF and ATT from multi-PLD ASL MRI. We highlighted an avenue for quality control via network uncertainty heuristics and demonstrated the potential to generalize to a range of multi-PLD schedules. By simplifying and improving efficiency of the ASL workflow, this deep learning-based pipeline is a step toward wider adoption of multi-PLD ASL MRI.

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DATA AVAILABILITY STATEMENT

Python code used to train the models, evaluate performance, and generate the figures presented in this manuscript will be available at https://github.com/nluciw/Deep-learning_for_mPLD-ASL. The TensorFlow-based models and weights will also be available.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

Figure S1 Network uncertainty heuristics were estimated through Monte Carlo dropout. Both images were acquired at 1.5T. Panel (A) shows a higher-quality estimate and (B) shows a lower-quality estimate.

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